

Seeing Directions with Closed Eyes: Exploring Eyelid-Transmitted Light as Visual Feedback

Shiryu Tanabe
University of Yamanashi
Kofu, Japan
g25tk020@yamanashi.ac.jp

Won-Du Chang
Pukyong National University
Busan, Republic of Korea
chang@pknu.ac.kr

Masahiro Toyoura
University of Yamanashi
Kofu, Yamanashi, Japan
mtoyoura@yamanashi.ac.jp

Kentaro Go
University of Yamanashi
Kofu, Japan
go@yamanashi.ac.jp

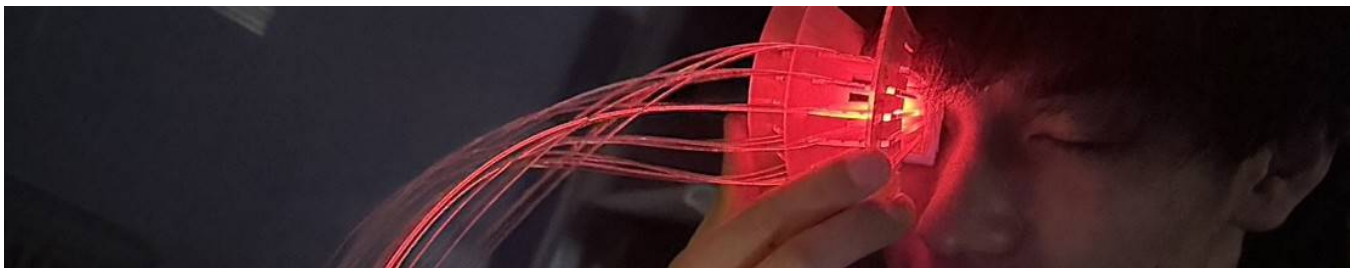


Figure 1: Using the proposed eyelid-transmitted light prototype under closed-eye conditions.

Abstract

Human-computer interaction typically assumes the availability of open-eye visual feedback, even in eyes-free interaction. However, visual processing is not entirely suppressed when the eyes are closed: light transmitted through the eyelids can evoke perceptual responses. This paper explores eyelid-transmitted light as an emerging visual feedback modality for closed-eye interaction. We present an early prototype that projects localized light stimuli onto the eyelid to convey directional information, and report an empirical probe examining whether users can interpret such cues. In a study with sixteen participants, performance in 2-, 4-, and 8-direction choice tasks exceeded chance level, with errors biased toward spatially adjacent directions. Response times increased with the logarithm of the number of alternatives, suggesting structured decision processes under degraded visual input. We invite discussion on the design space, perceptual constraints, and implications of using eyelid-transmitted light as a complementary feedback channel for closed-eye interaction.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; *Interaction techniques*; Empirical studies in HCI; Accessibility.



This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License.

CHI EA '26, Barcelona, Spain

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3799104>

Keywords

closed-eye interaction, eyelid-transmitted light, visual feedback, spatial perception, user study

ACM Reference Format:

Shiryu Tanabe, Masahiro Toyoura, Won-Du Chang, and Kentaro Go. 2026. Seeing Directions with Closed Eyes: Exploring Eyelid-Transmitted Light as Visual Feedback. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3799104>

1 Introduction

Human-computer interaction (HCI) typically assumes the availability of open-eye visual feedback. This assumption does not hold in visually constrained situations, such as intentional eye closure, including medical settings where eye opening cannot be assumed and entertainment experiences that deliberately restrict vision to enhance immersion. While eyes-free interaction has relied mainly on auditory and haptic modalities, these channels are limited in conveying spatial layouts and directional information. Importantly, visual processing is not fully eliminated with closed eyes: light transmitted through the eyelids can evoke perceptual responses [1], activate early visual pathways [2], and support gaze-independent cognitive processing [4].

In this work, we investigate an interaction technique based on eyelid-transmitted light projection to provide visual feedback under closed-eye conditions. Through a user study, we examine whether users can reliably perceive and respond to spatially distributed light cues and consider the implications for extending visual interaction beyond conventional displays.

2 Related Work

Imagraph demonstrated the potential of eyelid-transmitted light as a novel perceptual medium, focusing primarily on artistic and subjective experiences rather than functional interaction [7]. Building on this work, we investigate whether such stimuli can support systematic interaction.

To characterize user performance in closed-eye directional tasks, we refer to the Hick-Hyman law as a descriptive framework for choice behavior, which predicts a logarithmic increase in reaction time with the number of alternatives [3, 5] and has been widely used in HCI research [8], despite ongoing debate regarding its applicability in real-world settings [6].

The present study extends Imagraph toward a functional interaction paradigm by examining whether structured choice tasks using eyelid-transmitted light can elicit organized and interpretable responses under closed-eye conditions.

3 Interaction Design

This study proposes an interaction paradigm that conveys directional cues via eyelid-transmitted light under closed-eye conditions. The primary goal is to examine whether such stimuli can support reliable directional interaction when conventional visual feedback is unavailable. In addition, we observe response time characteristics as a secondary measure to characterize the decision-making process elicited by this form of feedback.

The system presents localized light stimuli on the eyelid, to which participants respond by selecting a discrete direction using a joystick. Stimulus positions are defined relative to the eyelid, providing a stable stimulus-response mapping independent of gaze direction or external visual context. Stimulus luminance is individually adjustable to ensure clear perception and user comfort.

This minimal design establishes a framework for evaluating how directional cues delivered through eyelid-transmitted light are perceived and for assessing the feasibility of closed-eye visual interaction.

4 Experimental Method

To examine the perceptual feasibility of eyelid-transmitted light as a directional feedback channel, we conducted a controlled laboratory study focusing on basic choice performance under closed-eye conditions.

Participants: Sixteen participants (13 male, 3 female; age: $M = 24.7$, $SD = 5.2$, range: 22–43) with normal or corrected-to-normal vision participated in the experiment. All participants provided informed consent prior to participation.

Apparatus: A custom prototype system was developed to present eyelid-transmitted visual stimuli. The system consisted of a WS2812-based LED matrix coupled with 1 mm diameter optical fibers, each guiding light to a localized position on the participant's eyelid. In each trial, only a single light point was activated.

An eyepiece fabricated from laser-cut acrylic plates ensured stable alignment between the optical fibers and the eyelid. Participants held the device with one hand and positioned it against either the left or right eye, freely chosen and kept constant throughout the experiment. Responses were made using a joystick held in the other hand.

Red light was employed because of its superior transmission through eyelid tissue and its stable perceived brightness under eye closure [9]. Stimulus luminance was individually adjusted prior to the experiment to the lowest level that allowed clear discrimination without discomfort.

Task and Conditions: Participants reported the perceived direction of the light stimulus under closed-eye conditions. Three choice conditions were tested using a within-subjects design: 2, 4, and 8 directional alternatives. Each direction was presented four times per participant in randomized order.

The 2-alternative condition included left and right directions; the 4-alternative condition included up, down, left, and right; and the 8-alternative condition additionally included diagonal directions. Directional options were defined relative to an eyelid-centered reference frame.

Prior to each condition, stimulus positions were individually calibrated so that all directions were clearly distinguishable. Calibration was repeated as necessary to maintain reliable discrimination.

Procedure: At the beginning of each trial, a single light stimulus was presented at one of the calibrated positions and remained illuminated until a response was made. Participants were explicitly instructed to respond as quickly and as accurately as possible immediately after stimulus onset using the joystick. They were also instructed to keep their eyes closed and to avoid intentionally moving their eyes during each trial.

Response time was measured from stimulus onset to the first registered joystick input. For analysis, responses exceeding 3 seconds were treated as timeouts. Immediately after responding, participants rated their subjective confidence on a 5-point Likert scale. No feedback regarding response correctness was provided.

Measures: Three dependent variables were recorded: response accuracy, response time, and subjective confidence. Joystick input angles were discretized into directional categories with a dead zone threshold applied to prevent unintended inputs; only the valid response in each trial was analyzed.

These measures were used to evaluate whether eyelid-transmitted visual feedback enables reliable directional discrimination and to examine whether response times vary with the number of alternatives in a manner consistent with the Hick-Hyman law.

Data Analysis: Response accuracy and response time were analyzed.

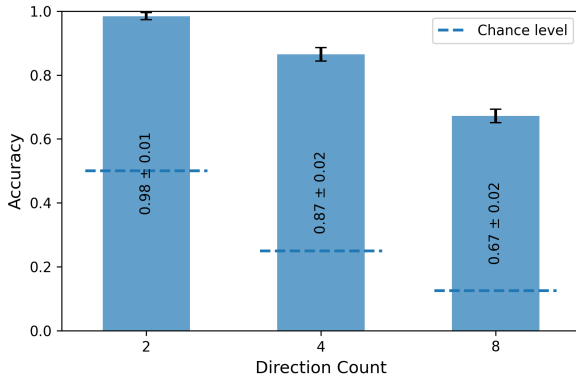
Response accuracy was analyzed using generalized estimating equations (GEE) with a binomial distribution and logit link function. The number of directional alternatives was entered as a $\log_2$ -transformed predictor, and an exchangeable working correlation structure was specified to account for repeated measures. A generalized linear mixed model was considered but not adopted due to unstable fitting; therefore, GEE was used as a population-averaged approach.

To assess whether accuracy exceeded chance level, post-hoc one-sided tests compared mean accuracy against chance performance ($p_{\text{chance}} = 1/N$) for each condition.

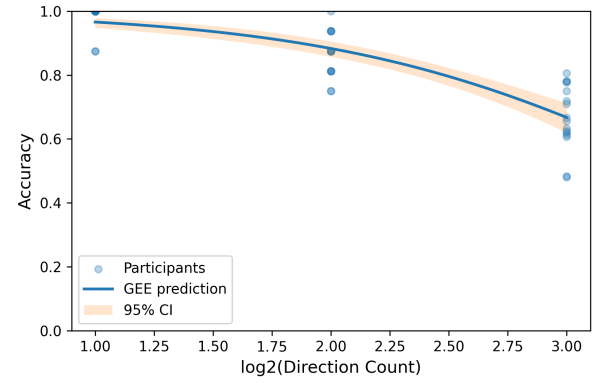
Response time was analyzed using a linear mixed-effects model with log-transformed response times:

$$\log(RT) \sim \log_2(N) + (1 + \log_2(N) \mid \text{participant}).$$

All statistical tests used an alpha level of .05.



(a) Mean accuracy and chance level



(b) Population-averaged accuracy (GEE)

Figure 2: Response accuracy under closed-eye conditions. (a) Mean accuracy for the 2-, 4-, and 8-direction conditions; dashed lines indicate chance ($1/N$) and error bars show standard errors. (b) Population-averaged accuracy as a function of $\log_2(\text{direction count})$ estimated by GEE; points show participant means, the line the model prediction, and the shaded area the 95% confidence interval.

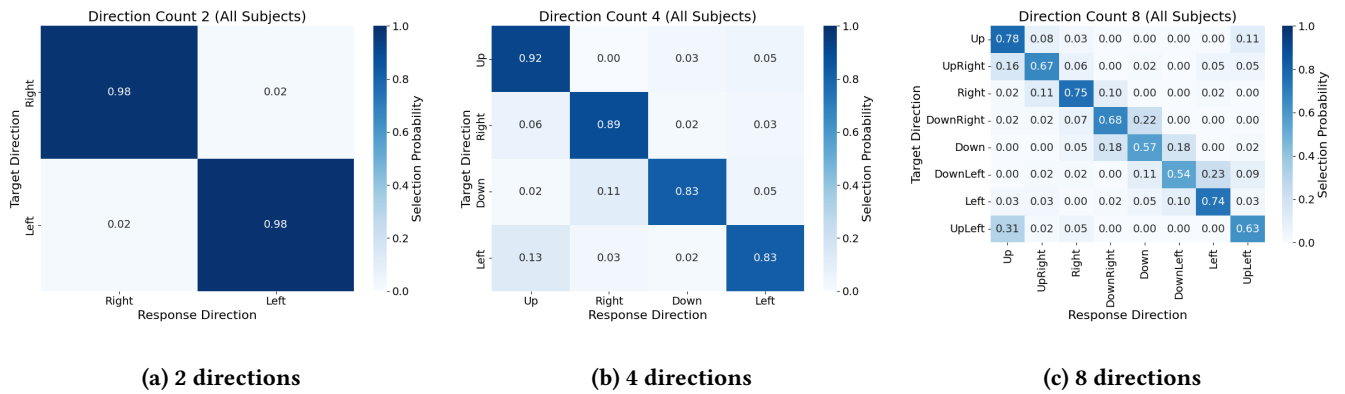


Figure 3: Confusion matrices for the 2-, 4-, and 8-direction conditions. Errors are concentrated in spatially adjacent directions as the number of alternatives increases.

5 Results

5.1 Response Accuracy

Figure 2 shows response accuracy under closed-eye conditions. Mean accuracy decreased with increasing numbers of response alternatives but exceeded chance level ($1/N$) in all conditions.

Generalized estimating equations (GEE) revealed a significant negative effect of $\log_2(N)$ on response accuracy ($\beta = -1.33$, $SE = 0.14$, $z = -9.65$, $p < .001$). Post-hoc comparisons confirmed that accuracy was significantly higher than chance for all conditions ($N = 2, 4, 8$; all $p < .001$).

5.2 Error Patterns Across Direction Conditions

Confusion matrices (Figure 3) showed that errors were systematically biased toward spatially adjacent directions in the 4- and 8-direction conditions, indicating structured perceptual confusions rather than random guessing.

5.3 Response Time and Hick–Hyman Law

Figure 4 illustrates mean response time as a function of $\log_2(N)$. Response time increased monotonically with the number of alternatives.

The linear mixed-effects analysis showed a significant positive effect of $\log_2(N)$ on response time ($b = 0.322$, $p < .001$). Although individual differences in intercepts and slopes were observed, the fixed-effect estimate was positive with a narrow 95% confidence interval $[0.245, 0.398]$, indicating a consistent trend across participants.

These results demonstrate that response time increased in proportion to the logarithm of the number of alternatives. While the limited number of levels prevents a definitive test of the Hick–Hyman law, the observed pattern was consistent with its predictions and did not contradict it.

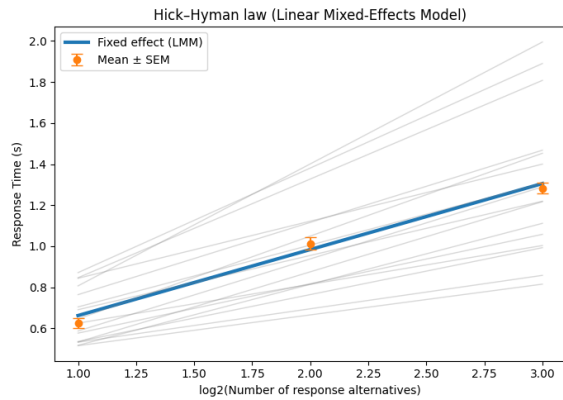


Figure 4: Response time as a function of $\log_2(N)$. Points show the grand mean for each condition with standard error bars. The thick line indicates the fixed-effect regression from the linear mixed-effects model, and thin lines show subject-specific predictions.

6 Discussion

This study examined whether eyelid-transmitted light can function as a usable visual feedback channel for structured interaction under closed-eye conditions. The results are interpreted in terms of perceptual organization, decision processes, and interaction design implications.

6.1 Eyelid-Transmitted Light as Structured Visual Information

Across all conditions, response accuracy exceeded chance level, indicating that participants extracted directional information from eyelid-transmitted light despite strong optical attenuation and spatial diffusion. This suggests that even highly degraded visual input can support categorical spatial judgments.

Errors were not uniformly distributed: confusions were systematically biased toward spatially adjacent directions, particularly in the 4- and 8-direction conditions. This pattern indicates that participants formed an organized but coarse internal spatial representation that preserved neighborhood relationships rather than precise locations.

6.2 Decision Processes under Closed-Eye Visual Feedback

Response times increased monotonically with the logarithm of the number of alternatives. Although the limited number of choice levels prevents a formal test of the Hick-Hyman law, the observed trend was consistent with information-theoretic models of choice behavior.

Notably, this scaling emerged despite the absence of open-eye vision and the degraded nature of the sensory input. This suggests that eyelid-transmitted visual feedback engages decision mechanisms similar to those used in conventional visual tasks, rather than relying on ad hoc strategies.

Together with the accuracy results, these findings indicate that participants actively categorized spatial alternatives rather than merely detecting the presence of light.

6.3 Implications for Closed-Eye Interaction Design

The results suggest that eyelid-transmitted light can support interactions involving a limited set of discrete alternatives, with accuracy decreasing and response time increasing as choice complexity grows. Error patterns indicated systematic confusions between spatially adjacent directions, suggesting that users formed a coarse spatial organization of the stimuli and that adjacency may be an important consideration when mapping functions to directional positions.

6.4 Limitations

Several limitations should be considered when interpreting the results.

First, although participants were instructed to keep their eyes still, eye position was neither measured nor controlled. Because the perception of eyelid-transmitted light can vary with eyeball orientation, unmonitored eye movements may have affected stimulus perception and response variability.

Second, the limited number of directional alternatives restricts conclusions regarding the functional form predicted by the Hick-Hyman law; thus, the response time analysis should be interpreted as descriptive.

Finally, the use of a joystick reduces ecological validity for hands-free interaction scenarios. Accordingly, the present study should be understood as demonstrating perceptual feasibility rather than a complete closed-eye interaction loop.

7 Conclusion and Future Work

This paper explored eyelid-transmitted light as an emerging visual feedback modality for closed-eye interaction. An empirical probe using simple directional choice tasks showed that participants could discriminate spatial light cues beyond chance level, with systematic confusions biased toward spatial adjacency and response times increasing with choice complexity. These observations suggest that even highly degraded visual input can support organized spatial categorization under closed-eye conditions.

Rather than presenting a finalized interaction technique, this work highlights eyelid-transmitted light as an underexplored design space and invites discussion on how visual feedback should be conceptualized when vision is degraded. Future work will investigate tighter integration with closed-eye input techniques, such as eye-movement-based interaction, and explore alternative spatial encodings and learning effects.

Acknowledgments

This research was partially supported by JSPS KAKENHI Grant Numbers 24K15633, 24K15029, and 22H00549. We are grateful to all participants who took part in our study.

References

- [1] Katsuhisa Ando and Daniel F. Kripke. 1996. Light attenuation by the human eyelid. *Biological Psychiatry* 39, 1 (Jan. 1996), 22–25. doi:10.1016/0006-3223(95)00109-3
- [2] Nicholas G. Cicero, Michaela Klimova, Laura D. Lewis, and Sam Ling. 2024. Differential cortical and subcortical visual processing with eyes shut. *Journal of Neurophysiology* 132, 1 (2024), 54–60. doi:10.1152/jn.00073.2024
- [3] William Edmund Hick. 1952. On the Rate of Gain of Information. *Quarterly Journal of Experimental Psychology* 4, 1 (1952), 11–26. doi:10.1080/17470215208416600
- [4] Han Jeong Hwang, Valeria Y. Ferreria, Daniel Ulrich, Tayfun Kilic, Xenofon Chatziliadis, Benjamin Blankertz, and Matthias Treder. 2015. A Gaze Independent Brain-Computer Interface Based on Visual Stimulation through Closed Eyelids. *Scientific Reports* 5 (2015), 15890. doi:10.1038/srep15890
- [5] Ray Hyman. 1953. Stimulus Information as a Determinant of Reaction Time. *Journal of Experimental Psychology* 45, 3 (1953), 188–196. doi:10.1037/h0056940
- [6] Wanyu Liu, Julien Gori, Olivier Rioul, Michel Beaudouin-Lafon, and Yves Guiard. 2020. How Relevant is Hick's Law for HCI?. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. Association for Computing Machinery, New York, NY, USA, 1–11. doi:10.1145/3313831.3376878
- [7] Goki Muramoto. 2022. Imagraph. In *ACM SIGGRAPH 2022 Art Gallery (SIGGRAPH '22)*. Association for Computing Machinery, New York, NY, USA, Article 9, 1 pages. doi:10.1145/3532837.3534947
- [8] Robert W. Proctor and Darryl W. Schneider. 2018. Hick's law for choice reaction time: A review. *Quarterly Journal of Experimental Psychology* 71, 6 (2018), 1281–1299. doi:10.1080/17470218.2017.1322622
- [9] Hideki Sakai. 2022. Perception of brightness when the eyes are closed. *Color Research and Application* 48, 1 (2022), 63–68. doi:10.1002/col.22832